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E1.
$$\begin{cases} \dfrac{du}{dt} = u^2 v & u(0) = 1 \\[2ex] \varepsilon \dfrac{dv}{dt} = -3uv + 3 & v(0) = 0 \end{cases}$$

$\varepsilon \to 0$:
$$-3uv + 3 = 0$$
$$\begin{cases} v = \dfrac{1}{u} \\[2ex] \dfrac{du}{dt} = u^2 \cdot \dfrac{1}{u} = u \end{cases}$$

$$u(t) = C e^t \qquad u(0) = C = 1$$
$$u(t) = e^t$$

soluzione esterna:
$$\begin{cases} \tilde{u}(t) = e^t \\ \tilde{v}(t) = \dfrac{1}{e^t} \end{cases}$$

soluzione interna:

$$\tilde{u}(t) = e^{t} + \frac{1}{1-t/\varepsilon^2} - L_u$$

$\boxed{\tau = t/\varepsilon}$ SOLUZIONE INTERNA:

$$\frac{du}{d\tau} = \frac{du}{dt}\cdot\frac{dt}{d\tau} = \varepsilon\frac{du}{dt} \qquad \Big| \qquad \frac{dV}{d\tau} = \frac{dv}{dt}\cdot\frac{dt}{d\tau} = \varepsilon\frac{dv}{dt}$$

$$\begin{cases} \frac{1}{\varepsilon}\frac{du}{d\tau} = u^2 v \\ \frac{dV}{d\tau} = -3uv+3 \end{cases} \rightarrow \begin{cases} \frac{du}{d\tau} = \varepsilon u^2 v & u(0)=1 \\ \frac{dV}{d\tau} = -3uv+3 & V(0)=0 \end{cases}$$

$$\varepsilon \to 0 : \begin{cases} \frac{du}{d\tau} = 0 \\ \frac{dV}{d\tau} = -3V+3 \end{cases} \rightarrow u_0(\tau) = 1$$

$B = 3$
$A = -3$

$$V(\tau) = V_0(\tau) + V_p(\tau)$$
$$= ke^{A\tau} + \frac{B}{A} = ke^{-3\tau} - 1$$

a $V(0)=0 \longrightarrow ke^{-3\cdot 0} - 1 = 0 \rightarrow k=1$

$$V(\tau) = e^{-3\tau} + 1$$

$\cancel{v = u^2(1-e^{-3\tau})}$

$\cancel{u = 0}$

MERGING CONDITIONS:

V:

$$\lim_{t \to 0} \frac{1}{e^t} = 1$$

$$\lim_{\tau \to \infty} e^{-3\tau} + 1 = 1$$

U:

$$\lim_{t \to 0} e^t = 1$$

$$\lim_{\tau \to \infty} 1 = 1$$

$$\tilde{u}(t) = e^t + 1 - 1 = e^t$$

$$\tilde{v}(t) = e^{-t} + e^{-3t/\varepsilon} - 1$$

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